

10. General Conclusion and Outlook

10.1 - General Conclusion

This thesis asks how a FinOps data platform can transform FOCUS cost-and-usage records into reliable products and provide governed analysis for decision support. The project has produced an independent synthetic-data generator, a local DuckDB and Streamlit prototype, and a Databricks/GCP implementation with separate DEV and PROD processing paths. The available evidence supports different conclusions for each research question.

For reliability, the local validation confirms a reproducible source history and reconciled January 2025 analytical outputs. The validated Daily corpus contains 608 files through August 2026; the unchanged monthly billing corpus contains eighteen months through June 2026. The versioned contract, audit model and month-replacement logic specify how the cloud platform should preserve these relationships. Historical DEV and PROD processing is reported in the Belgium plan, but complete run-level artifacts are still needed to establish the cloud result independently. The full Daily-to-monthly-close cycle has not yet been validated in the thesis evidence package.

For governed self-service, the local application executes against published datamarts, which demonstrates a functioning analytical interface in that environment. It does not establish that managers or analysts can use the cloud application within verified access boundaries. Identity-based filtering, direct SQL permissions and task completion require separate tests. The reported professional problem also concerns the time taken to change an existing dashboard; that process has not been measured.

For FinOps decision support, the local datamarts expose cost perspectives, service concentration and allocation coverage. These figures come from synthetic data. The high share of `Unknown` cost-center allocation shows that a reconciled charge can still be weak evidence for accountability. Cost differences cannot be called savings without validated charge semantics and a comparable baseline. The platform's own operating cost is likewise defined methodologically but has not been quantified from a complete Databricks and GCP bill.

The project therefore establishes an auditable engineering and evaluation basis for FinOps analysis, with a locally tested subset of its intended capabilities. Further cloud, access and business validation determines how far the same design can support governed organizational use.

10.2 - Learning Outcomes and Application of Master's Coursework

The project connects several areas of the master's curriculum to one data-engineering problem. Schema design and data-quality methods appear in the versioned FOCUS contract and its Bronze-to-Silver gate. Dimensional modeling appears in the Gold fact, dimensions and datamarts. Distributed processing concepts inform the Databricks implementation, whereas DuckDB provides a separate local validation setting. The distinction between these settings is methodologically significant: code that is structured for cloud execution and an observed cloud run are different forms of evidence.

The project also places analytical measures in a business context. The negative contracted-to-effective difference in the local January dataset required checking the sign and meaning of an indicator before interpreting it as a benefit. The large `Unknown` cost-center share required distinguishing numerical reconciliation from useful allocation. These examples illustrate the application of data modeling and quality concepts to FinOps questions; they do not establish that the student personally acquired a particular skill or completed a named course.

JSON fields, tags and metadata are examples of semi-structured information used in the project. Their use does not imply that a NoSQL database was deployed. Forecasting remains a possible extension; no machine-learning result is reported. The application of coursework can be assessed against the implemented components and their validation limits, while personal learning outcomes require separate first-person evidence.

10.3 - Future Work

The immediate technical extension is an evidence-backed cloud validation of the complete Daily-to-monthly-close cycle. It requires a DEV-to-PROD promotion, replay of an already processed file, reconciliation against a monthly billing candidate, and recovery from a controlled interruption. Retained run identifiers, table outputs and costs would also permit a controlled Serverless and Classic comparison. These tests address specific gaps identified in chapters 7 and 8.

Governed use first requires the planned Cloud Run persona experiment to test denial by default and role-and-scope filtering across every dashboard query. A production extension would replace the synthetic selector with a validated Google IAP identity, synchronize authorized users and groups with the enterprise directory, review entitlements periodically and retain sanitized access logs. The backend service principal should remain read-only and limited to the serving products and security metadata. If users later receive direct analytical access, Unity Catalog views, row filters or equivalent controls must protect the same scopes outside the dashboard. Representative manager and analyst tasks on authorized data would

then test whether the products answer the intended questions and whether the change workflow improves compared with the recorded baseline.

Forecasting could be investigated after a target quantity or cost measure, forecast horizon and simple reference method are defined. Chronological validation on the synthetic history would test only behavior under the generator's assumptions. Evidence from representative real history would be needed before describing operational forecasting value.